

Ryan Rahman

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EDUCATION

University of Waterloo

Waterloo, ON

B.A.Sc. Mechatronics Engineering, Artificial Intelligence Option

Accelerated M.A.Sc. Electrical and Computer Engineering, Robotics Specialization

TECHNICAL SKILLS

Robot Learning: VLA Models, Flow Matching, Diffusion Policies, World Action Models, Imitation Learning, RL (SAC, PPO, DSRL, RECAP), Multimodal Fusion

Robotics: ROS2, Isaac Sim/Lab, MuJoCo/MJLab, Pinocchio (IK), Real-Time Control, SLAM, Manipulation, Teleoperation, Sim-to-Real

Languages and Tools: Python, C++, JavaScript, SQL, PyTorch, JAX/Flax, HuggingFace, vLLM, CUDA, Docker, Linux, Weights & Biases, Git

EXPERIENCE

Founding Research Engineer

Sep. 2026 – Present

Preload (YC F26)

San Francisco, CA

- Founding research hire **scaling EMG data collection** and fusing surface-EMG with **egocentric vision** for **manipulation policy pretraining**, building the data engine and multimodal training stack from scratch

Student Researcher

Aug. 2026 – Present

Amazon | University of Waterloo

Remote

- Research **sequential decision models for human-robot last-mile delivery**, modelling handoff and dispatch as a learned policy over partially observed state
- Built **simulation and offline-evaluation infrastructure** to benchmark learned policies against classical planning baselines prior to fleet deployment

Research Engineer

Apr. 2026 – Sep. 2026

AXIBO

Cambridge, ON

- Developed a **Mixture-of-Horizons action head** for $\pi_{0.5}$ flow-matching VLAs with disagreement-based adaptive replanning, raising full-task success from **10% to 70% on espresso making** and **60% to 90% on towel folding** over fixed-horizon baselines across **100 rollouts per task**
- Built the **asynchronous inference path** behind our smoothest and most accurate rollouts: **10 Hz** policy inference decoupled from **30 Hz** control, overlapping action chunks fused by **temporal ensembling** (newest four, age-weighted), and latency compensated by shifting each chunk's start backwards rather than discarding its leading actions, giving **zero buffer starvation** across 50–250 ms policy latency
- Developed a **proprioceptive–tactile subgoal estimator** for a **65-DoF** bimanual platform, trained on DTW-aligned subtask segmentation, reducing action error **47% on visually ambiguous frames**
- Fine-tuned VLA policies in **noise space (DSRL)** with a $\sim 500\text{K}$ -parameter SAC agent steering a frozen flow policy, improving manipulation success **30% relative** to the imitation baseline; also built $\pi_{0.6}$ RECAP offline-RL and Proxy Policy Steering pipelines
- Investigated **where tactile sensing should enter manipulation policy networks**, staging ablations from raw force/torque concatenation to vision-co-attending prefix tokens to a **force-aware mixture-of-experts** decoder

Robotics Engineer

Dec. 2025 – Feb. 2026

Syncere AI

Palo Alto, CA

- Collected **~ 450 expert teleoperation demonstrations per week** for imitation learning and $\pi_{0.5}$ -style VLA policies, emphasizing precise end-effector control for long-horizon bimanual manipulation
- Ran policy-pipeline experiments on **EE-only trajectory supervision**, action-chunk scaling, depth/fisheye fusion, and tactile encodings to improve multimodal representations and rollout stability
- Investigated **scalable demonstration formats** for robot policy learning, comparing end-effector pose trajectories against richer proprioceptive observations for trajectory replay and policy supervision